

Clinical Risk Assessment Report: Patient 41

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 13.4% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.61x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which reduced the patient's absolute risk by 2.3%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

- Squamous Cell Carcinoma Antigen (SCC_Ag)

Value: 0.6 [Bottom 18%] (+1.3%)

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.
- Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT)

Value: 2.79 [Top 2%] (+0.9%)

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.
- Tumor Pixel Correlation (CT) (GLCM_Correlation.CT)

Value: 2.05 [Top 1%] (+0.9%)

Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

3. Protective / Mitigating Factors (Reducing Risk)

- Large Zone Emphasis (CT) (GLZLM_LGZE.CT)

Value: 28.7 [Top <1%] (-2.3%)

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.
- Patient Age (Age)

Value: 72.0 [Top 18%] (-2.1%)

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.
- Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET)

Value: 0.2 [Top 22%] (-1.2%)

Points to continuous, elongated bands of high-density or highly active tumor tissue.
- Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT)

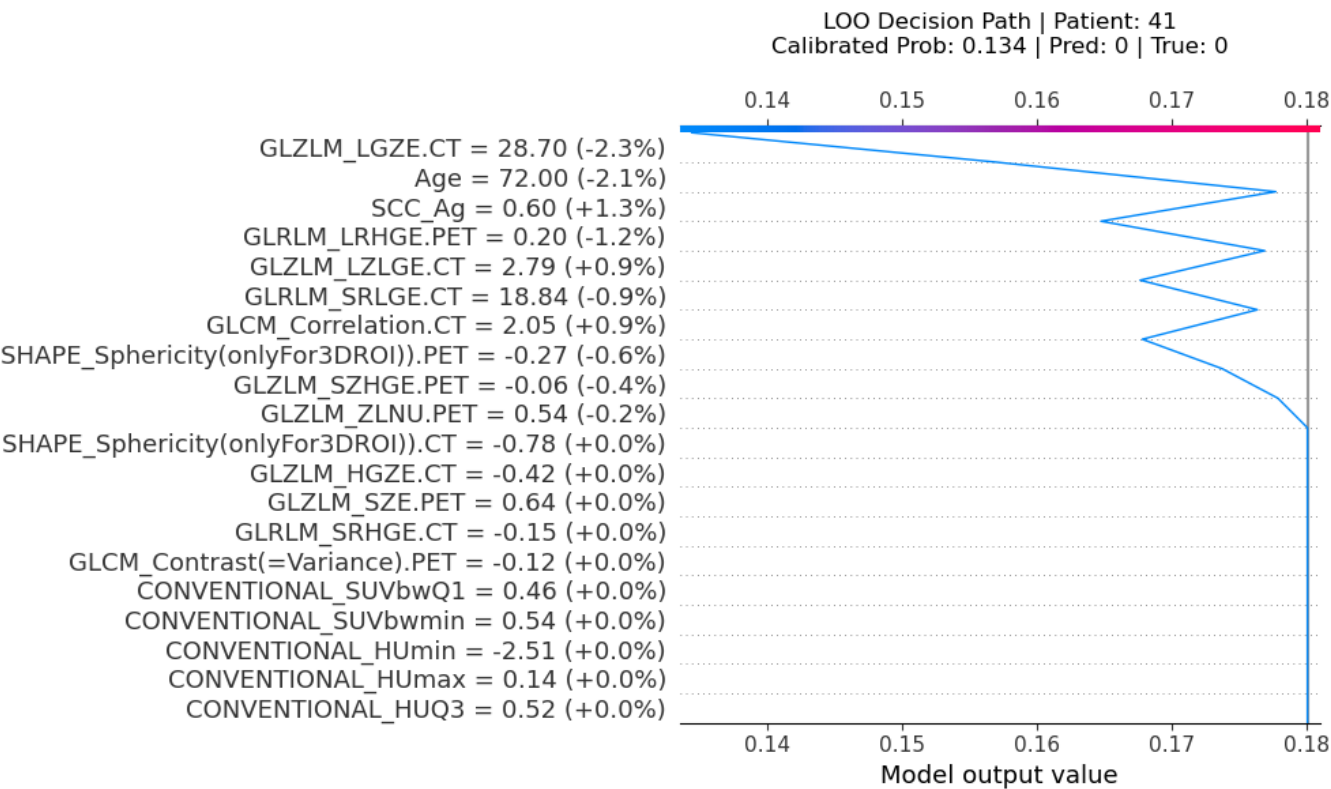
Value: 18.84 [Top <1%] (-0.9%)

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.
- Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET

Value: -0.27 [Avg (31th %ile)] (-0.6%)

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

